

Do **not** write solutions on this page.

11. [Maximum mark: 17]

The plane Π_1 has equation $x + 2y + z = 0$ and the plane Π_2 has equation $x - y - 2z = 0$.

The acute angle between the planes Π_1 and Π_2 is θ .

(a) Show that $\theta = 60^\circ$.

[6]

A third plane Π_3 is perpendicular to both Π_1 and Π_2 .

The unique point of intersection of all three planes is the point $R(5, -5, 5)$.

(b) Find the Cartesian equation of Π_3 .

[4]

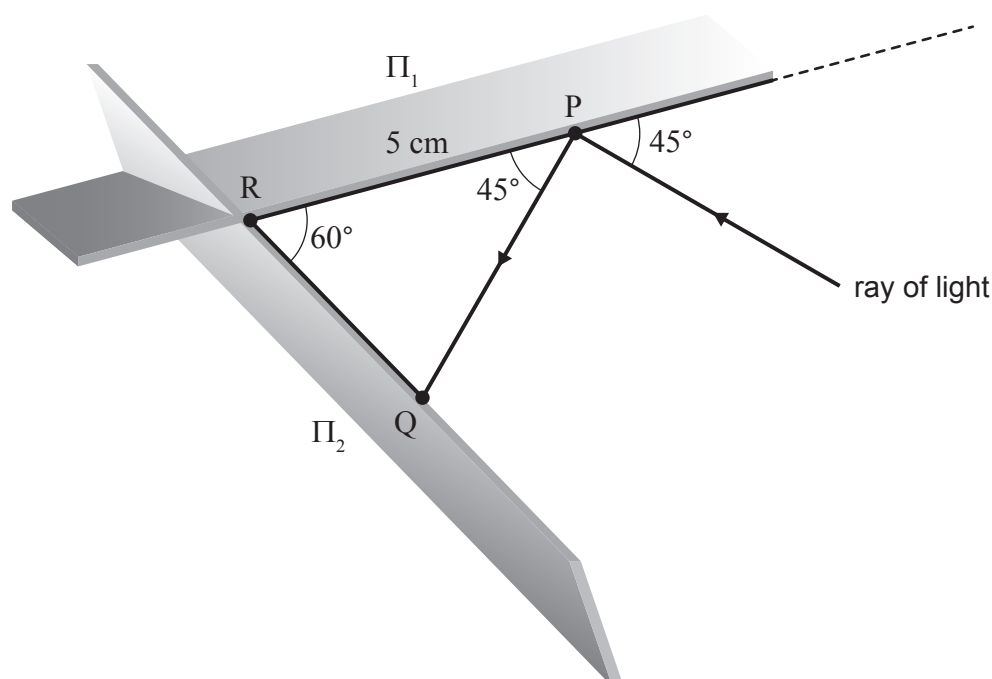
Each of the planes Π_1 and Π_2 contains a mirror.

A ray of light is directed towards the mirror in Π_1 . The ray of light forms an angle of 45° with Π_1 and meets it at the point P .

The ray of light is then reflected towards the mirror in Π_2 , and meets Π_2 at the point Q . The points P and Q are contained in Π_3 .

It is given that $PR = 5$ cm.

This information is shown on the following diagram.



(This question continues on the following page)



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(Question 11 continued)

(c) (i) Using an appropriate compound angle identity, show that $\sin 75^\circ = \frac{\sqrt{2} + \sqrt{6}}{4}$.

(ii) Find QR, giving your answer in the form $p(\sqrt{q} - 1)$ cm where $p, q \in \mathbb{Z}$. [7]

